

Cash Generation under AAOIFI Debt Limits: An Exploratory Backtest of Halal FCF Quality, 2023–2024

Rao Abdul Hadi*¹

¹Monterey Finance, Halal Quant Research Lab

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Abstract

This paper asks whether, after Islamic business and leverage screens have been applied, firms that convert a large share of sales into free cash flow hold up as a Halal equity book when owned in proportion to size. I start from the S&P 500, drop banned activities and names that fail AAOIFI financial-ratio tests, keep the top half of the remaining list by free-cash-flow (FCF) margin, and cap-weight the survivors. In the two calendar years 2023–2024 the book returned 42.4% compound annual growth, against 25.9% for the S&P 500 (SPY) and 31.1% for a Halal large-cap ETF (SPUS). The live rule is cash generation, not cheapness. A prior FCF-to-enterprise-value sort missed Apple and Nvidia and lost to SPUS in the same window, so that rule was dropped. The resulting portfolio is a concentrated Halal mega-cap quality and technology book: technology plus communication services averaged about 74% of weight. Quintiles of FCF margin inside the Halal universe are not monotonic, though the top two buckets lead. A robustness sleeve that drops the jumpiest fifth of names still beats both benchmarks, at a lower CAGR. The sample is short, the FCF data come from Yahoo annual statements, the rule was rewritten after seeing 2023–2024, and I do not estimate a sector-neutral twin. The numbers should be read as an exploratory backtest, not as a forecast or a live-return target.

Keywords: FCF margin; free cash flow; AAOIFI; Sharia-compliant equities; quality factor; backtest.

JEL classification: G11, G12, Z12.

Code: Replication notebook and figures are in the Monterey Finance repository, <https://github.com/regional-specter/Monterey-Finance>, under Research/papers/01-fcf-ev/.

1 Introduction

Cash generation is a quality signal. Free cash flow is cash from operations minus capital spending. FCF margin is that cash divided by sales. A high margin means the business turns a large share of revenue into cash after reinvestment. That is not the same as cheapness. A firm can produce a lot of cash relative to sales and still look expensive on yield if the market has already paid up for it.

The quality literature has long treated profitability and cash generation as priced characteristics (Novy-Marx, 2013; Fama and French, 2015; Asness et al., 2019). Islamic equity screens add a second filter. AAOIFI financial-ratio limits cap interest-bearing debt, cash, and receivables relative to market value, and a separate activity screen removes banks, conventional insurance, alcohol, tobacco, gambling, weapons, and similar businesses (AAOIFI, 2020; Derigs and Marzban, 2008).

*Corresponding author. Email: raoabdulhadi952@gmail.com.

The economic hope is that those caps already strip out highly indebted firms, so that among what remains, cash generation is a cleaner quality signal.

That hope is not automatic. Sharia screens change the sector mix. They tend to cut financials and some utilities, and they leave more weight in technology and other operating businesses. Any backtest that looks good after those screens may simply be riding the same mega-cap platform names that dominated 2023 and 2024.

The original design for this study ranked Halal names by cheapness: FCF divided by enterprise value, equal-weighted in the top fifth. That book lost to SPUS in 2023–2024. Apple and Nvidia passed the Halal screens but looked expensive on FCF/EV, so they were not bought, while SPUS is a broad, cap-weighted Halal basket that owned the mega-cap winners. The live rule is therefore different. After banned businesses and the AAOIFI tests, keep names with positive FCF whose FCF margin sits in the top half of that month, and own them in proportion to market cap. Cheapness is not used to pick or weight names. FCF/EV is stored only as a diagnostic.

That rewrite used the same 2023–2024 window, so the headline results are partly in-sample. I keep that fact in view throughout.

On headline returns the cash-generation book beat both SPY and SPUS. It also took a slightly deeper drawdown than SPY, it is heavily concentrated in a handful of platform names, and FCF-margin quintiles inside the Halal universe are not a clean stair-step. I treat the result as evidence that a Halal cash-generation rule can be implemented and was not a backtest bug, not as proof that AAOIFI debt caps amplify a quality premium across cycles.

The rest of the paper is organized as follows. Section 2 describes the universe, screens, and factor definitions. Section 3 states the portfolio rules. Section 4 reports live-window performance, calendar years, drawdowns, and the quintile sort. Section 5 looks at alpha, beta, and sector mix. Section 6 covers turnover, purification, data limits, and the in-sample rule change. Section 7 concludes.

2 Data, screens, and factor definitions

2.1 Universe

The starting list is the current S&P 500 (503 tickers). A one-time activity screen removes 64 names, leaving 439 sector-eligible stocks. Financial-ratio screens then re-run at every month-end on filings that would have been known as of that date. I use `filed_date` when it is available, and otherwise the report date plus a 90-day publication lag.

The requested price sample is 1 January 2019 through 31 December 2024. Yahoo annual cash-flow statements do not go back far enough for a 2019 start. The first date with usable FCF metrics is 30 November 2021. Performance statistics begin on the first day the strategy actually holds stocks, 3 January 2023, and run through 30 December 2024 (501 trading days). Days before that first invested date are omitted, not filled with zeros.

Yahoo had no in-sample price history for Q, SNDK, and FDXF (names listed after 2024). Those names are skipped.

2.2 Screening standard

The financial screens follow AAOIFI ratios, not the Dow Jones Islamic Market (DJIM) thresholds (AAOIFI, 2020; Derigs and Marzban, 2008). The debt rule in this test is total debt below 30% of trailing 24-month market cap, together with the library’s cash and receivables screens. Compliance is a hard constraint. A name that fails on a rebalance date is not held.

SPUS (SP Funds S&P 500 Sharia Industry Exclusions) is the Halal index comparison. It is a

cap-weighted, industry-exclusion ETF with its own reconstitution calendar. It is not identical to a pure AAOIFI financial-ratio index, so beating SPUS is not the same as beating “the” Sharia market. This run does not include a DJIM-tracking book. SPY is the all-stock benchmark.

Table 1 summarizes the design choices.

Table 1: Universe, screens, and sample.

Item	Choice
Target factor	FCF margin = free cash flow / sales
Starting universe	Current S&P 500 list (503 tickers)
After banned-business screen	439 tickers (64 removed)
Screening standard	AAOIFI financial ratios, not DJIM
Halal benchmark	SPUS (SP Funds S&P 500 Sharia Industry Exclusions)
All-stock benchmark	SPY (S&P 500)
Sample requested	1 Jan 2019 – 31 Dec 2024
Live window used for stats	3 Jan 2023 – 30 Dec 2024 (501 trading days)

2.3 FCF margin

Free cash flow is cash from operations minus capital spending. FCF margin is

$$\text{FCF margin} = \frac{\text{FCF}}{\text{Sales}}.$$

Names must have positive FCF to enter. FCF comes from yfinance annual cash-flow statements, lagged 90 days. That is a pragmatic proxy until audited cash-flow tags live in the research library. Restatements and tag mapping are not checked here.

I do not claim that the AAOIFI debt cap creates a quality premium by itself. The 2023–2024 sample is too short and too technology-heavy to separate the screen from the sector mix.

3 Portfolio rules

Rules are applied at month-end with no look-ahead. There is no intra-month stop-loss and no single-name position cap beyond cap-weighting.

3.1 Pipeline at each month-end

1. Start from the sector-eligible S&P 500 list (439 names).
2. Drop names that fail the AAOIFI debt, cash, or receivables screens on filings known as of that date.
3. Compute FCF margin as FCF divided by sales. Require $\text{FCF} > 0$.
4. Keep the top half of that Halal list by FCF margin (`KEEP_QUANTILE` = 0.50).
5. Cap-weight by market cap. No equal-weight, no inverse-volatility, and no optimizer.
6. Repeat at month-end (`REBALANCE_FREQ` = ME).

Table 2: Entry, exit, and risk limits.

Rule	Specification
Entry	Halal, FCF > 0, FCF margin in the top half that month
Exit	Sold at the next month-end if the name fails AAOIFI, loses positive FCF, or drops out of the top half
Intra-month stop-loss	None
Position cap	None beyond cap-weight
Cash when thin	Skip the month if holdings < 20
Robustness twin	Same screen, then drop the jumpiest fifth of those picks (252-day volatility)

If fewer than 20 names survive (MIN_HOLDINGS), that month is skipped and the book stays in cash. Cash months are treated as missing, not as a 0% return. Statistics start on the first invested date, not on the 2019 calendar start.

At the last rebalance (31 December 2024), Apple was 13.4% of the book, Nvidia 11.9%, Microsoft 11.1%, and the two Alphabet share classes together were about 16.8%. That is a concentrated mega-cap book by construction. A 5% or 8% name cap would change both risk and 2023–2024 return. I leave that for a follow-up rather than rewriting the live rule after seeing the weights.

3.2 Holdings through time

Median live book size is 124 names (minimum 20, maximum 130). Early months sit on the 20-name floor. Coverage then jumps and stays in the 120–130 range through 2024. At a typical month the funnel is roughly 419 names with fundamentals, 233 Halal names with positive FCF, and 116 actually bought.

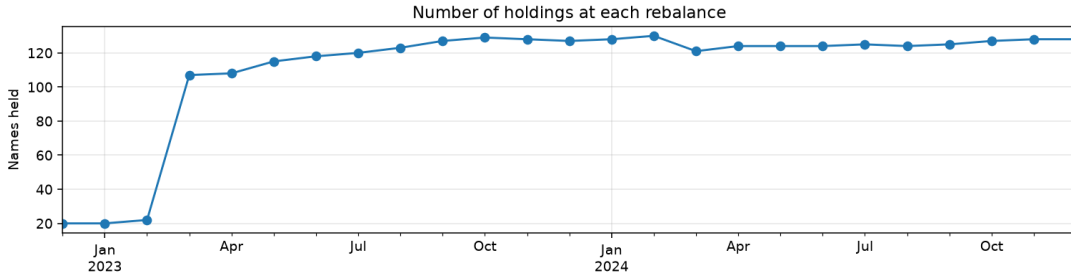


Figure 1: Number of names held at each month-end. The 20-name floor binds in late 2022 and early 2023. The book then becomes a broad, cap-weighted quality slice of the Halal S&P list.

Sharia status and the FCF-margin rank are re-checked every month. A held name that fails is exited at that rebalance. It is not left in the book pending a discretionary review.

On 31 December 2024 the watchlist looked as follows. Held names included AAPL, NVDA, MSFT, GOOGL, META, and AVGO: all Halal-ok, all in the top half by FCF margin. Apple’s FCF yield was only 2.8%, which would have looked expensive on the old cheapness rule, with a 27.8% FCF margin, which is why it is kept under the live rule. Amazon ranked 224 of 255 on FCF margin and Tesla ranked 234 of 255. The screen keeps about 128 names, so both sat in the bottom half and were not bought. Lilly had an FCF margin of −9.2% that month and was not bought. Lilly was held in 12 of 38 metric months when FCF was positive and high enough.

Amazon and Tesla were never in the live book (0 of 38 months). That is the cash-generation screen

working as written, not a data accident at the last date. Microsoft and Nvidia show missing-snapshot months in the autopsy: Yahoo or SEC files were absent for some month-ends, so the name was skipped that month even if it would have passed.

GOOG and GOOGL were both held as separate lines. Cap-weighting two share classes of the same company inflates Alphabet relative to a single-listing index. That is a construction error to fix in a follow-up.

4 Empirical performance

All headline numbers use the live window only: 3 January 2023 through 30 December 2024. Sharpe and Sortino use a 2% risk-free rate. Maximum drawdown is peak-to-trough on the growth curve. Yahoo’s short FCF history is why the book does not go live in 2019.

4.1 Headline results

Table 3: Live-window performance, 3 Jan 2023 – 30 Dec 2024.

Book	CAGR	Vol.	Sharpe	Sortino	Max DD	Calmar
Halal FCF quality	42.43%	16.13%	2.15	3.10	−11.92%	3.56
Quality (drop high vol)	34.26%	14.45%	1.97	2.85	−9.39%	3.65
S&P 500 (SPY)	25.93%	12.82%	1.71	2.50	−9.97%	2.60
Halal (SPUS)	31.11%	15.07%	1.74	2.58	−11.46%	2.71

The quality book beat both benchmarks on CAGR, Sharpe, Sortino, and Calmar. It took a slightly deeper drawdown than SPY (−11.9% versus −10.0%) and a similar one to SPUS. Dropping the jumpiest fifth of the quality names cut CAGR to 34.3% but also cut volatility and drawdown. Calmar stayed similar (3.65 versus 3.56). That overlay still beat SPY and SPUS. It is a robustness check, not a second live product.

These CAGRs are two-year annualized figures from a mega-cap bull market. They are not a forecast of 42% a year.

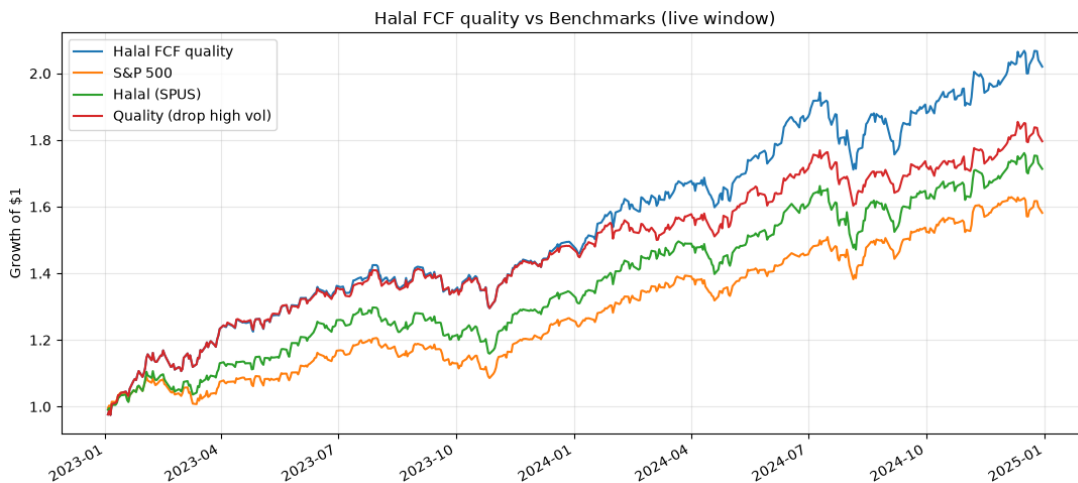


Figure 2: Growth of \$1 in the live window. The quality book finishes near \$2.00, the low-vol overlay near \$1.80, SPUS near \$1.70, and SPY near \$1.60.

4.2 Calendar years

Table 4: Calendar-year total returns inside the live window.

Year	Halal FCF quality	S&P 500	Halal (SPUS)
2023	49.12%	26.18%	34.20%
2024	35.47%	25.34%	27.67%

The quality book led in both years: by about 23 points versus SPY and 15 versus SPUS in 2023, and by about 10 and 8 points in 2024. There is no down year in this sample.

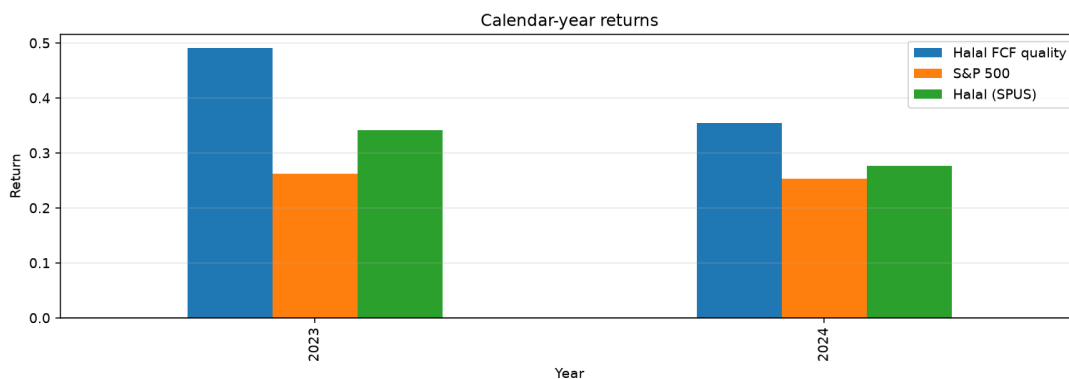


Figure 3: Calendar-year returns. The quality book is the tallest bar in both 2023 and 2024.

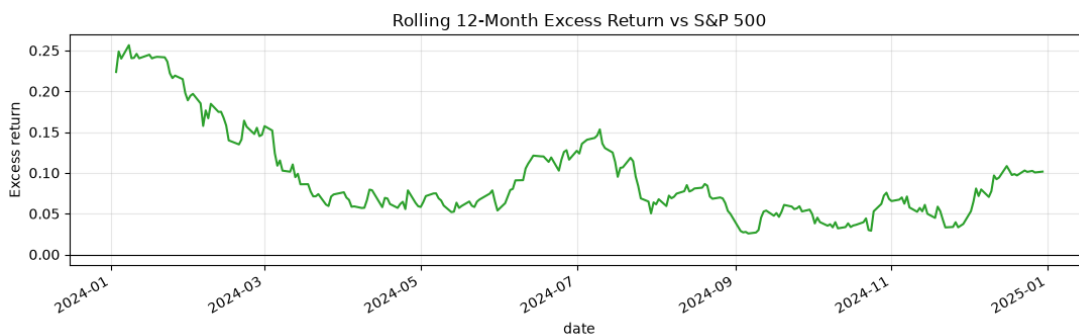


Figure 4: Rolling 12-month excess return versus SPY. Excess stayed positive through the live window, starting near +25% in early 2024 and ending near +10%.

4.3 Drawdowns

The quality book and SPUS dip together in late 2023, mid-2024, and late 2024. SPY's mid-2024 drawdown is shallower. That pattern is consistent with a Halal mega-cap technology book, not with a defensive quality sleeve.

4.4 Does higher FCF margin actually pay?

Inside the same Halal universe I cap-weight five FCF-margin buckets. Q1 is the highest cash generation and Q5 is the lowest. The live strategy is roughly the top half (near Q1–Q2), not a single

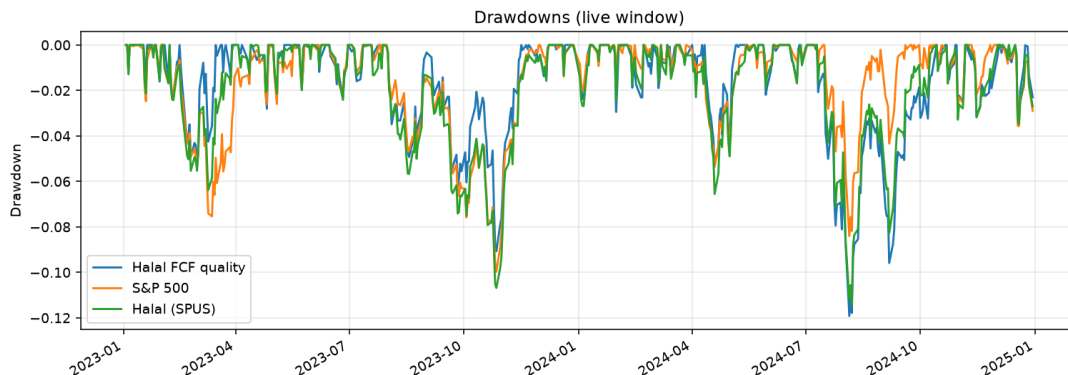


Figure 5: Drawdowns in the live window. Paths track SPUS more closely than SPY. The quality book’s worst mark (about -11.9%) is in mid-2024, when Halal and technology names sold off harder than the full S&P 500.

quintile.

Table 5: Cap-weighted FCF-margin quintiles inside the Halal universe, live window. Q1 is highest FCF/sales.

Bucket	CAGR	Vol.	Sharpe	Sortino	Max DD	Calmar
Q1	37.44%	19.34%	1.64	2.35	-15.14%	2.47
Q2	35.76%	16.34%	1.83	2.48	-10.03%	3.57
Q3	27.39%	16.86%	1.40	2.26	-13.08%	2.09
Q4	11.62%	14.96%	0.68	1.14	-14.16%	0.82
Q5	31.08%	15.07%	1.74	2.60	-11.74%	2.65

Q1 and Q2 are the two best CAGRs, which supports keeping the top half. The pattern is not perfectly monotonic. Q5 beat Q3 and Q4. Q2 had the best Sharpe and Calmar. The Q1 minus Q5 annualized spread was 5.5%, with Q1 ahead of Q5 on 51.3% of days. The cumulative spread ended near $+10\%$ after peaking around 1.28.

Cash generation helped in this window. It did not line up as a clean, stair-step factor.

5 Attribution

Section 4 shows that the book won on total return. This section asks whether that win looks like stock picking or like a hidden sector bet.

5.1 Alpha, beta, and tracking error

Daily excess returns of the quality book are regressed on each benchmark (CAPM-style, 2% risk-free rate). Alpha is annualized.

Table 6: Halal FCF quality versus each benchmark, live window.

Benchmark	Alpha (ann.)	Beta	R^2	Tracking error	Info. ratio
S&P 500	9.56%	1.15	0.83	6.86%	1.87
Halal (SPUS)	7.97%	1.02	0.91	4.95%	1.71

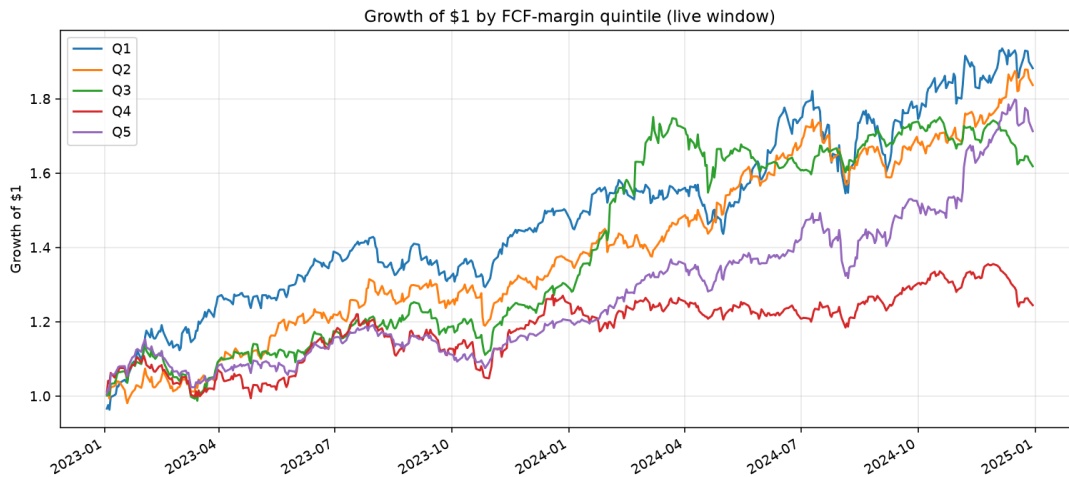


Figure 6: Growth of \$1 by FCF-margin quintile. Q1 and Q2 lead. Q4 lags. Q5 is not last.

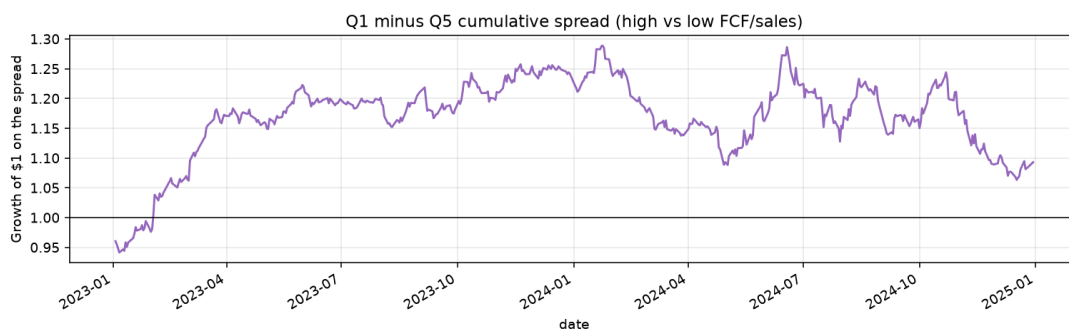


Figure 7: Cumulative Q1 minus Q5 spread. After early 2023 the line stays above 1, but it is noisy and faded from a peak near 1.28 to about 1.10 by the end of 2024.

Versus SPY the book is a bit more aggressive ($\beta = 1.15$) with 9.6% leftover annualized alpha and 6.9% tracking error. Versus SPUS it is almost a parallel book: $\beta = 1.02$, $R^2 = 0.91$, tracking error 5.0%, still 8.0% alpha. This is an SPUS-like Halal large-cap book with extra mega-cap technology, not an independent style that moves on its own.

The information ratios (1.87 versus SPY, 1.71 versus SPUS) look large because the sample is two strong years. They will shrink if a 2022-style year is added.

5.2 Sector exposure

Average month-end weights of the cap-weighted quality book are compared with an equal-weight basket of the Halal S&P universe. Sector labels follow the activity screen, not official GICS ETF weights.

Table 7: Average sector weights. Delta is quality book minus equal-weight Halal universe.

Sector	Quality book	Halal universe (EW)	Delta
Technology	51.5%	19.1%	+32.4%
Communication services	22.8%	5.5%	+17.3%
Healthcare	12.0%	13.4%	-1.5%
Energy	3.5%	4.8%	-1.3%
Consumer defensive	3.2%	6.4%	-3.2%
Industrials	2.9%	14.4%	-11.5%
Consumer cyclical	2.8%	11.6%	-8.8%
Financial services	2.7%	6.2%	-3.4%
Basic materials	1.3%	4.6%	-3.3%
Real estate	1.1%	6.6%	-5.6%
Utilities	0.0%	7.1%	-7.1%

Technology plus communication services average 74.3% of the book. Utilities are empty. Industrials, cyclicals, real estate, and utilities are the main underweights. An equal-weight Halal universe is much more balanced. Cap-weighting high-FCF-margin names loads the book onto the largest software and platform companies.

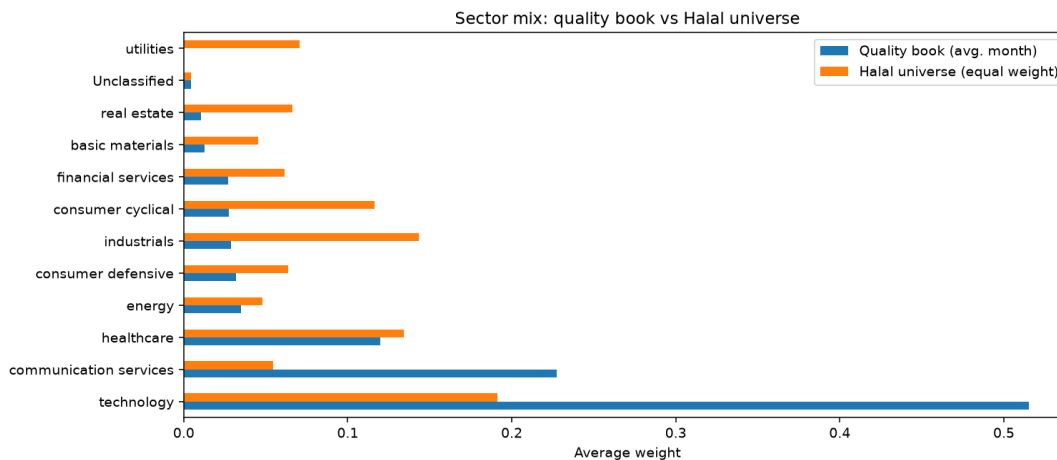


Figure 8: Sector mix: quality book versus equal-weight Halal universe. The quality book is a technology and communication overweight, not a diversified Halal S&P clone.

5.3 What can and cannot be isolated

The leftover alpha versus SPUS is still positive after a $\beta \approx 1$ fit, so the result is not only a restatement of SPUS. SPUS itself is already Halal and technology-tilted in this window, and this book is more concentrated in the same names. Without a sector-neutral twin I cannot say how much of the 8% SPUS alpha is FCF margin versus even more Apple, Nvidia, Microsoft, Alphabet, and Meta than SPUS already owned.

The fair reading is that stock selection inside a Halal screen produced a mega-cap quality and technology book, and that book paid in 2023–2024.

6 Limitations and implementation friction

6.1 Turnover, slippage, and net CAGR

Turnover is one-way: half the ℓ_1 weight change between consecutive month-end books. Trading-cost drag assumes 10 basis points per one-way turn (a round-trip of 20 bp on the traded sleeve).

Table 8: Implementation friction on the live quality book.

Item	Value
Rebalances with holdings	25
Average names held	110.9
Annualized one-way turnover	99.9%
Assumed cost per one-way trade	10 bp
Estimated annual trading-cost drag	0.10%
CAGR gross of trading costs	42.43%
CAGR net of trading costs	42.22%

Cap-weighting a broad top-half screen keeps monthly turnover low after the initial build, often a few percent a month, with a spike near March 2023 when the book expanded off the 20-name floor. Annualized one-way turnover of 99.9% looks high because that first 100% build and the March jump are included. Cost drag at 10 bp is only 10 bp a year, not large enough to change the headline. Real slippage on mega-caps should be tighter than 10 bp. On the smaller names in the tail it could be wider. I do not model market impact, borrow, or taxes.

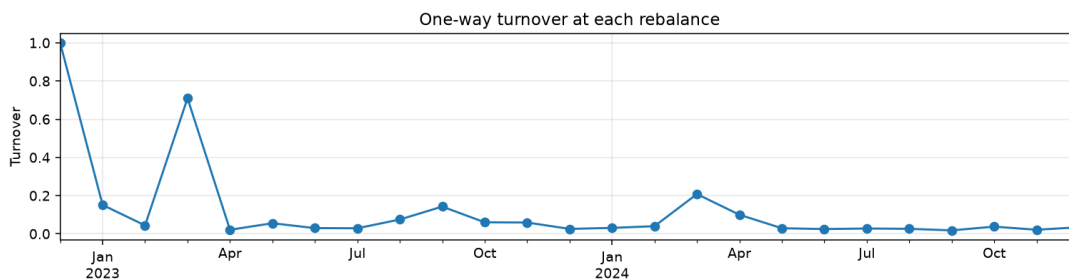


Figure 9: One-way turnover at each rebalance. The first point is 100% (book construction). After the March 2023 expansion, monthly turnover is usually well below 10%.

6.2 Purification drag

Impure income that may require purification is reported, not ignored. Using the library’s dividend-purification helper on names that appeared in the book, there were 450 dividend events while held, and annualized purification drag is about 0.01%. That is negligible next to 42% CAGR. Missing interest-income tags are a gap, not evidence of zero drag. The 0.01% figure is a lower bound on the names and fields available here, not a Sharia opinion that purification is optional.

6.3 Point-in-time compliance

Screens are applied at each rebalance. Failed names are sold then. There is no intra-month compliance monitor. If a name crossed the 30% debt line mid-month, the backtest would still hold it until month-end. That is optimistic relative to a strict daily AAOIFI process and pessimistic relative to indices that reconstitute quarterly.

Sector exclusions use the current S&P list and current activity labels, not the 2019 membership file. Names that were not in the S&P in 2023 but are in it now can appear too early. Names that left the index can be missing. That is survivorship and look-ahead in the universe, even though the filings themselves use `filed_date`.

6.4 Data and construction issues

Yahoo annual cash-flow statements typically store only a few years. The first metrics date is 30 November 2021 and the first invested date is 3 January 2023, so the 2019–2022 request is mostly unused. yfinance FCF is a pragmatic proxy. Restatements and tag mapping are not audited here.

GOOG and GOOGL are both weighted, so Alphabet is overweight versus a single-listing index. There is no position cap, which means three names can be a third of the book.

The more serious issue is the rule change. Cheap FCF/EV was scrapped because it lost to SPUS in this same window. 2023–2024 is therefore partly in-sample for the live FCF-margin rule.

6.5 What this paper does not show

There is no 2022, no 2008, no rate-hike recession, and no hold-out after a rule freeze. Quintiles are not monotonic. Sector mix explains a large part of the path. Beating SPUS in a two-year mega-cap boom is interesting. It is not evidence that AAOIFI debt caps amplify a quality premium across cycles, which was the original question.

7 Conclusion

A cap-weighted book of AAOIFI-screened S&P 500 names with FCF margins in the top half beat SPY and SPUS in 2023–2024. The product shape is coherent: Halal screens, a cash-generation cut, and cap-weighting. It was not a construction bug. It is also a mega-cap technology and platform book, the FCF-margin quintiles are noisy, and the live rule was chosen after a cheapness sort failed in the same sample.

The idea is worth a second paper. It is not ready to promote to a live mandate. That follow-up would need a frozen rule, SEC cash-flow tags with a longer history, point-in-time S&P membership, a single listing per issuer, a sector-capped twin, and at least one down year. If the sector-capped twin still beats SPUS, the quality claim has more to stand on. If it does not, the 42% CAGR in this sample is mostly the 2023–2024 technology tape.

Reproduction. Run `code.ipynb` in this paper’s folder with `FAST_MODE=False` and dates 2019-01-01 to 2024-12-31. Figures are stored alongside the notebook. The public repository is <https://github.com/regional-specter/Monterey-Finance>.

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